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Research paper

Ancestry prediction efficiency of the software *GenoGeographer* using a z-score method and the ancestry informative markers in the Precision ID Ancestry Panel

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ABSTRACT

We compared the efficiency of the freely available software *GenoGeographer* that includes a z-score based analysis with that of a naïve method based on the maximal likelihoods of 164 of the 165 ancestral informative markers (AIM) that are included in the commercially available kit Precision ID Ancestry Panel from Thermo Fisher Scientific. The AIM profiles were obtained by investigations with the Precision ID Ancestry Panel in our laboratory and from SNP data in the literature and publically available databases. We established eight well-defined AIM reference population data sets from 3603 AIM profiles. Six reference populations with profiles from multiple populations (Sub-Saharan Africa, North Africa, Middle East, Europe, South/Central Asia, East Asia), and two populations with individuals with admixed ancestry (Somalia and Greenland). By means of *GenoGeographer* and naïve calculations of the maximal likelihoods, 566 AIM profiles from individuals that were not included in the reference populations and expected to belong to one of the eight reference populations were tested. An initial standard z-score based test with *GenoGeographer* demonstrated that 22.4% of the individuals could not be assigned to any of the reference populations. Among the remaining 77.6% of the individuals, 83.6% were assigned to the reference population that was concordant with the specified populations of origin of the individuals, 8.2% had ambiguous assignments because they could belong to both the specified population of origin and one or more of the other populations, and 8.2% were assigned to a reference population that was discordant from the specified population of origin. A naïve assignment based on the maximal likelihood resulted in 78.1% concordant and 21.9% discordant assignments. The results demonstrate that the z-score analysis with *GenoGeographer* can reduce the error rate with a factor of almost three compared with that of the naïve estimation based on the maximal likelihoods of the AIM profiles. The Precision ID Ancestry Panel is a useful kit for the assignment of ancestry of the eight investigated populations that included two admixed populations. More AIMs with better discrimination and more data on the distribution of AIMs in relevant populations are needed to improve the efficiency of genogeographic prediction with AIMs on a worldwide basis.

1. Introduction

In crime and identification cases, it may be important to obtain information about ancestry and/or geographic origin of the investigated individual(s). This is of special interest in cases in which DNA from the perpetrator(s) is available but no suspect is identified, and no match is found in crime DNA databases. Similarly, ancestry information is of value in identification of missing persons and disaster victims. Ancestry, the geography of human populations, and genetic polymorphisms are closely associated with each other [1–5]. Harrison [6] used the term 'genogeographic markers' for genetic markers with alleles that are differently distributed among various populations and, thus, may reveal information of the affiliation of individuals to populations, which is often associated with a certain geographic location. The term ancestry informative markers (AIMs) is often used for such genetic markers. Cavalli-Sforza et al. [7] published an extensive review of the information about ancestry that could be obtained by investigation of genetic markers at the time. Since then, highly informative AIMs have been identified (reviewed by Jobling et al. in 2014 [8]). Autosomal Short Tandem Repeats (STRs), which are presently the

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standard tool for human identification and relationship testing in forensic genetics, give information on genogeographic ancestry, although the information is limited [9]. Y-chromosome markers [10] and mitochondrial (mtDNA) sequence variation [11] give valuable information concerning paternal and maternal lineages. Autosomal markers – especially Single Nucleotide Polymorphisms (SNPs) – are the most frequently used ancestry markers in forensic genetics because a large number of SNPs have very different allele frequencies among populations [12–16]. Insertion-deletion polymorphisms (indels) may also be valuable AIMs [17–19]. Indels can be analysed with the same methods as those used for forensic genetic STR typing, e.g. PCR amplification and capillary electrophoresis, which is very convenient for forensic routine laboratories. However, the number and informative value of indels are less than those of SNPs [16,20].

Investigation of genogeographic ancestry is especially valuable if a relatively large number of markers (e.g. SNPs) can be investigated. In crime case investigations, it is important to be able to perform the investigations on very small amounts of DNA, i.e. less than 1 ng of DNA. Massively Parallel Sequencing – MPS – is a sensitive method that makes it possible to investigate a large number of SNPs, including AIMs, simultaneously with less than 1 ng of DNA [21,22]. Commercial kits including AIMs are available and are used for characterization of ancestry. We use the Precision ID Ancestry Panel (Thermo Fischer Scientific) that includes 165 genetic SNP markers [23,24]. The kit is one of the first commercially available kits developed to discriminate between the major ethnic groups from Sub-Saharan Africa, Europe, and Asia. The knowledge of the distribution of genetic markers in human populations is continuously increasing [13]. Databases with information on genogeographical markers have been established e.g. ALFRED: https://alfred.med.yale.edu [25,26], SNIPPER: http://mathgene.usc.es/snipper [27], and the Estonian Biocenter (http://evolbio.ut.ee) that includes data from Clemente et al. [28] and Pagani et al. [29].

Biostatistical predictive tools for forensic genetic use have been developed e.g. [30]. Most tools either (1) estimate the maximal likelihood of the AIM profile in various populations, and the population with the highest likelihood is considered the most likely population of origin or (2) infer the presence of distinct populations and assign individuals to the inferred populations or admixtures thereof (e.g. STRUCTURE [31,32]). The likelihoods of observing the AIM profile in two or more populations and the ratio(s) between the likelihoods may be calculated, offering the likelihood ratio as the weight of the evidence as recommended by the International Society for Forensic Genetics – ISFG [33,34]. However, the naïve maximum likelihood method is not optimal, because the likelihoods of the AIM profile may be very similar in two or more populations, thus making the conclusion uncertain and possibly misleading. Furthermore, none of the populations considered may be relevant, and thus the likelihood ratio may be misleading – even with high LR values. Therefore, statistical methods should be used to estimate (1) the certainty of the acceptance of an individual with a certain AIM profile as a possible member of a population and (2) the weight of the evidence taking into account the profile uncertainties when two or more population affiliations are possible.

Presently, AIM data are not available for all populations. Therefore, the databases may not include the appropriate reference populations leading to uncertain or misleading results of the ancestry estimation [13,23]. The fact that the likelihood is substantially larger in one population than in another does not necessarily prove that the AIM profile can be grouped together with the AIM profiles of a particular population. To handle this situation, we derived a likelihood ratio test, a z-test, by which it can be assessed if there is any reference population in the database that is sufficiently close to the "true" ancestry population of the individual with the investigated AIM profile [35]. If the null hypothesis that the AIM profile originates from a given population in the database is rejected for all reference populations in the database, no further calculation is performed. Otherwise, likelihood ratios that compare the probabilities of the evidence under hypotheses of different populations of origin are calculated. The method is implemented in the freely available software *GenoGeographer* [35].

The version of *GenoGeographer* that we used was not optimised for analysis of individuals with ancestry from two or more populations. In the meantime, the likelihood ratio test approach used in *GenoGeographer* was extended by allowing for profile admixture implying the profile of interest is a mixture of profiles from two populations [36], which will allow prediction of the ancestry of individuals of mixed ancestry.

The purpose with this study was to evaluate the efficiency of *GenoGeographer* compared to methods based on simple likelihood maximisation using a commercially available AIMS kit. Thus, it was not the intention to evaluate other specific applications or AIM panels.

We compared the efficiency of prediction of ancestry by means of (1) *GenoGeographer* and (2) the naïve method based on the maximal likelihood of the AIM profile. SNP data of the 165 SNPs in the Precision ID Ancestry Panel from our own AIM typing data [23,24] and from the literature, as well as databases allowed us to identify eight statistically significantly different metapopulations. The eight metapopulations were Sub-Saharan Africa, Somalia, North Africa, Europe, Middle East, South/Central Asia, East Asia, and Greenland. The metapopulation of origin of test individuals with known genogeographic origin were predicted with the *GenoGeographer*.

2. Materials and methods

2.1. Reference population data

A simulation study showed that for acceptable estimates of the approximate confidence intervals [37] of the likelihoods of the alleles in a population, at least 75 individuals per population were needed (data not shown). We investigated seven populations from Denmark, Greenland, Somalia, Morocco, Iraq, Iran, and Turkey with more than 75 individuals using the Precision ID Ancestry Panel [23,24] and unpublished). We supplemented these data with publically available data from the literature and databases as well as genotype data kindly provided by Professor Kenneth Kidd and his group, leading to a total of 3603 individuals from 45 geographic areas, populations, and subpopulations [38] (Supplementary Table 1). From the data, 36 populations that included 75 or more individuals typed for at least 148 of the 165 (> 90%) AIM SNPs were obtained and used for further analysis. The data for rs10954737 was excluded from the population data, because the genotype information was not available for all the individuals considered. The AIM profiles from the majority of populations with high degrees of admixture (e.g. the Hazara and the Peruvian populations) were excluded for further analysis.

We analysed the data from subgroups from the same region but with fewer than 75 individuals with *Genepop v. 4.7.0* [39] (http://kimura.univ-montp2.fr/~rousset/Genepop.htm) to determine if the distribution of allele- and genotypes of the AIM SNPs differed statistically significantly among the subgroups. The data from seven Tunisian subgroups were combined as a Tunisian group, because the SNP allele frequencies did not differ statistically significantly among each other ($P \geq 0.05$; z-score ≤ 1.64), and neither did data from two Russian and two Nigerian subgroups, respectively [13] (Supplementary Table 1). The proportion of genetic ancestry of each individual was inferred using the software STRUCTURE v. 2.3.4.21 [31,32]. The analysis was carried out using a burn-in series of 100,000 iterations, and 100,000 subsequent iterations were used to generate a sample from the posterior of size 100,000. The model with 'Admixture' and 'correlated allele frequencies' was considered. The number of assumed populations (K) was $K = 4$, as initial analyses demonstrated this was the most optimal K (data not shown). Three independent runs were carried out to check the consistency of results. The cluster membership coefficients, Q, generated with STRUCTURE were visualized with CLUMPP v.1.1.222 [40] (https://web.stanford.edu/group/rosenberglab/clumpp.html) and

Table 1
Assignment of population of the test individuals based on the AIMs results with the Precision ID Ancestry Panel and analysis with *GenoGeographer*.

Observation	Interpretation
No accepted population	No population of origin was assigned
One accepted population	The population was assigned
≥ 2 reference populations accepted	
- one reference population with statistically significantly higher likelihood than all other investigated populations (LR = 1 was not included in the confidence interval of the LR between the highest and the second highest population likelihoods; in popular terms: $P < 0.05$).	The population with the highest likelihood was assigned
- Two or more reference populations with likelihoods that were not statistically different (LR = 1 was included in the confidence interval of the LR between the highest and the second highest population likelihoods; in popular terms: $P \geq 0.05$).	Ambiguous, i.e. more than one possible population of origin

DISTRUCT v.1.1.23 [41] (https://web.stanford.edu/group/rosenberglab/distruct.html). The data of the individuals in each reference population were ordered in Excel (Microsoft) according to the proportions of the four major genetic contributions (K = 4).

Principal Component Analysis (PCA) was performed using an in-house script written in Python based on the software *sklearn.decomposition.PCA v. 0.16.1* [42]
(http://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html.

The geographical positions of the individuals were found with the software *latlong* (https://www.latlong.net).

2.2. AIM profiles of test individuals

Samples from 91 individuals from 67 different countries and self-declared ancestry were typed with the Precision ID Ancestry Panel (Thermo Fischer Scientific) as previously described [24]. The data were supplemented with previously obtained data of 13 individuals from Greenland [23]. The data were further supplemented with AIM profiles from 182 randomly selected individuals from 41 populations from Pakstis et al. [26,38] and K. Kidd (personal communication) not used in the reference population database. Information concerning some of the markers was missing, but at least 148 SNPs (> 90% of the total number of SNPs) were typed. A total of 402 AIM profiles with 161 of the 165 AIMS SNPs in the HID Precision Ancestry kit (rs735480, rs2835370, rs4411548, and rs1876482 were missing) were extracted from the study of Pagani et al. [29]. This study also includes data from Clemente et al. [28]. The data are publically available at the Estonian Biocenter (http://evolbio.ut.ee). VCF-files were downloaded, and the genotypes of 161 AIMS SNPs were extracted using the software R and the package *vcfR* [43].

AIM profiles from Siberian individuals were excluded for further analysis due to lack of genetic homogeneity among AIM profiles. AIM profiles of individuals from South America were excluded due to lack of relevant reference population data. Furthermore, individuals with less than 148 (< 90%) AIM SNPs typed were excluded. In total, 566 AIM profiles from individuals from 83 geographic areas were compared with those of the eight reference populations. Supplementary Table 2 summarises the numbers of test individuals in the various populations of origin specified in the various data sets.

2.3. Analysis of AIMs profiles with GenoGeographer

GenoGeographer is a publically available software that was developed for estimation of the geographic/population origin of individuals based on AIM profiles [35]. *GenoGeographer* can be accessed through http://genogeographer.org/. *GenoGeographer* was set up with default parameters, except that *exponential tilting* (importance sampling) was enabled.

The *GenoGeographer* software first tests if the investigated AIM profile can be grouped together with any of the reference populations in the database by means of the z-score test (https://en.wikipedia.org/wiki/Z-test). The test includes an estimation of the variance of the allele frequencies in the reference populations [37]. If the P-value of the AIM profile belonging to a reference population is < 0.05 ($P_{rejection} < 0.05$), *GenoGeographer* rejects the hypothesis that the AIM profile can belong to that reference population. The analysis is repeated for all reference populations resulting in a conservative assessment of the evidence in accordance with forensic practice. Thereafter, *GenoGeographer* calculates the likelihood ratios of the AIM profile in the various non-rejected reference populations. Finally, the likelihoods are sorted according to the values of the likelihoods. A likelihood ratio (LR) can be calculated in order to compare the likelihoods of the AIM profile in each of two reference populations. Several LRs may be calculated by selecting the relevant populations to be compared.

The AIM profiles of each of the 566 individuals were analysed with *GenoGeographer*. The p-value of the "population in database", the likelihood, and – if relevant – one or more likelihood ratios were calculated for each individual. The individuals were assigned to populations according to the criteria in Table 1.

2.4. Ethics

The work was approved by the Danish ethical committee (H-1-2011-081).

3. Results

3.1. Reference population database

We collected AIM profiles of 3603 individuals from 36 populations that included 75 or more individuals. All individuals were typed for at least 148 of the 165 (> 90%) AIM SNPs of the Precision ID Ancestry Panel. Supplementary Fig. 1 shows a map with the geographical areas of the populations included in the reference material. Analysis with STRUCTURE showed four major genetic components (K = 4): Sub-Saharan African, European, South/Central Asian, and East Asian. Fig. 1 shows a plot of the AIM profiles analysed with STRUCTURE and K = 4 with the proportions of the four major genetic components of each individual.

Six metapopulations and two highly admixed populations were identified. Greenlanders and Somalis are relevant to forensic genetic casework in Denmark (cf. below). Although both populations are admixed, we included them in the analysis. The six metapopulations plus the Greenland and Somali populations are hereafter called reference populations (Fig. 1). We named the reference populations according to the name of the geographical regions: Sub-Saharan Africa, Somalia, North Africa, Europe, Middle East, South/Central Asia, East Asia, and Greenland. The differences of the genotype frequencies among the eight reference populations were all statistically significantly different ($P < 10^{-6}$ for all comparisons, z-score test).

Supplementary Fig. 2 shows the principal component analysis (PCA) of the AIM profiles of the 3603 individuals in the eight reference populations and the 566 test individuals.

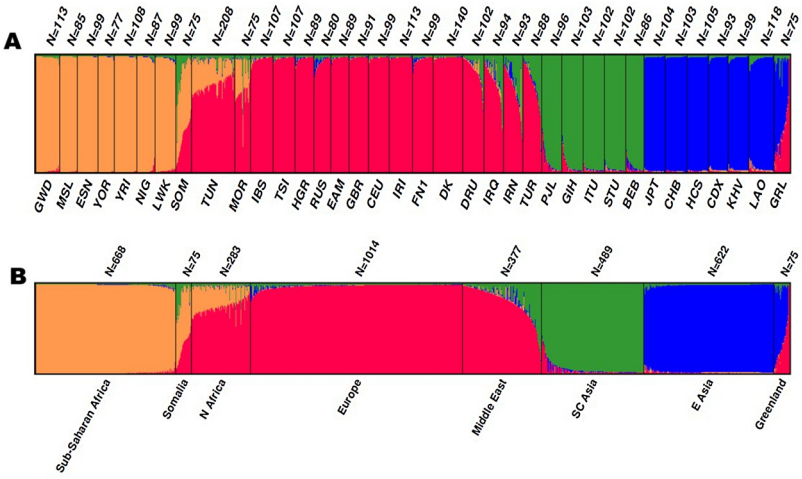


Fig. 1. A) DISTRUCT v.1.1.23 plot of the membership coefficients calculated by STRUCTURE *v. 2.3.4.21* for the AIM profiles of 3603 individuals from 36 geographic areas, populations, and sub-populations. B) DISTRUCT v.1.1.23 plot of the AIM profiles of the 3603 individuals grouped into the eight reference populations: Sub-Saharan Africa, Somalia, North Africa, Europe, Middle East, South/Central Asia, East Asia, and Greenland. The colours in both plots indicate the four major genetic components. Orange: Sub-Saharan African. Red: European. Green: South/Central Asian. Blue: East Asian. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.2. Comparison of results with GenoGenographer and the naïve method

In order to test if it was reasonable to assume that an individual could belong to one or more of the eight reference populations, the P-values for inclusion in each of the reference populations were calculated with *GenoGeographer*. Of the 566 test individuals, 127 (22.4%) could not be assigned to any of the reference populations, while 439 (77.6%) of the individuals were assigned to at least one reference population (Table 2). The assigned reference population was in concordance with the specified (expected) population of origin of the individual (concordant) for 367 (83.6%) of the individuals. In cases, in which more than one population were accepted, only assignments of populations with statistically significantly higher likelihoods ($P < 0.05$, Table 1) than others were considered concordant. A total of 36 assignments (8.2%) were considered ambiguous because the differences in likelihoods between the populations with the two highest likelihoods were not statistically significant. The assignment of population was not concordant with the specified population of origin of the individual (discordant) for 36 (8.2%) of the individuals.

If a naïve approach of assignment of population based on the highest likelihood was used, 442 (78.1%) individuals were concordantly assigned, while 124 (21.9%) individuals were discordantly assigned Table 2).

Fig. 2 shows the geographical positions of the 566 test individuals with the assignment result indicated by colour. The data represent areas and populations listed in Table 2. Large areas of the world and numerous populations were not represented.

Fig. 3 shows the distribution of the z-scores for the AIM profiles of all test individuals in each of the eight reference populations. The individuals belonging to the East Asian, Sub-Saharan African, Somali, and Greenlandic reference populations were clearly separated from individuals from other populations by the z-scores, while e.g. the z-score distribution of the European test individuals was very similar to those of the Middle East, the North African, and the South/Central Asian

Table 2
Assignment of population (or geographic region of origin) of 566 individuals by means of AIM profiles and *GenoGeographer*.

Expected population	Calculations based on z-score and confidence interval										Naïve calculation			
	Population assignment					Population assignment					Population assignment			
	Total	None[a]		One or more[b]		Concordant[c]		Ambiguous[d]		Discordant[e]	Concordant[f]		Discordant[g]	
	N	N	(%)	N	(%)	N	(%)	N	(%)	N	(%)	N	(%)	N	(%)
Sub-Saharan Africa	37	7	(18.9)	30	(81.1)	22	(73.3)	1	(3.3)	7	(23.3)	33	(89.2)	4	(10.8)
Somalia	22	5	(22.7)	17	(77.3)	17	(100.0)	0	(0.0)	0	(0.0)	21	(95.5)	1	(4.5)
North Africa	49	4	(8.2)	45	(91.8)	37	(82.2)	7	(15.6)	1	(2.2)	46	(93.9)	3	(6.1)
Europe	173	32	(18.5)	141	(81.5)	107	(75.9)	18	(12.8)	16	(11.3)	113	(65.3)	60	(34.7)
Middle East	52	5	(9.6)	47	(90.4)	41	(87.2)	5	(10.6)	1	(2.2)	46	(88.5)	6	(11.5)
South/Central Asia	85	33	(38.8)	52	(61.2)	38	(73.1)	5	(10.0)	9	(17.3)	47	(55.3)	38	(44.7)
East Asia	121	40	(33.1)	81	(66.9)	79	(97.5)	0	(0.0)	2	(2.5)	109	(90.1)	12	(9.9)
Greenland	27	1	(3.7)	26	(96.3)	26	(100.0)	0	(0.0)	0	(0.0)	27	(100.0)	0	(0.0)
Total	556	127	(22.4)	439	(77.6)	367	(83.6)	36	(8.2)	36	(8.2)	442	(78.1)	124	(21.9)

[a] *None:* The likelihood ratio tests of observing the AIM profile in each of the reference populations were all rejected (z-score > 1.64; $P < 0.05$, one-sided test); it was concluded that, for practical purposes, the AIM profile could not belong to any of the reference populations.

[b] *One or more:* The AIM profile could belong to one or more of the reference populations (z-score ≤ 1.64; $P \geq 0.05$ (one-sided test)).

[c] *Concordant:* The AIM profile could belong to one or more of the reference populations; in cases with two or more possible populations, the population with the highest likelihood was statistically significantly higher ($P < 0.05$) than that with the second highest likelihood. The assigned reference population was in concordance with the specified (expected) population of origin of the individual.

[d] *Ambiguous:* The AIM profile could belong to two or more populations; the difference in likelihoods between the two highest likelihoods was not statistically significant ($P > 0.05$).

[e] *Discordant:* The assigned population was not in concordance with the specified (expected) population of origin of the individual.

[f] *Concordant:* Assignment based on a naïve estimate of the population with the maximal likelihood of the AIM profile. The assigned population was in concordance with the specified (expected) population of origin of the individual.

[g] *Discordant:* Assignment based on a naïve estimate of the population with the maximal likelihood of the AIM profile. The assigned population was not concordant with the specified (expected) population of origin of the individual.

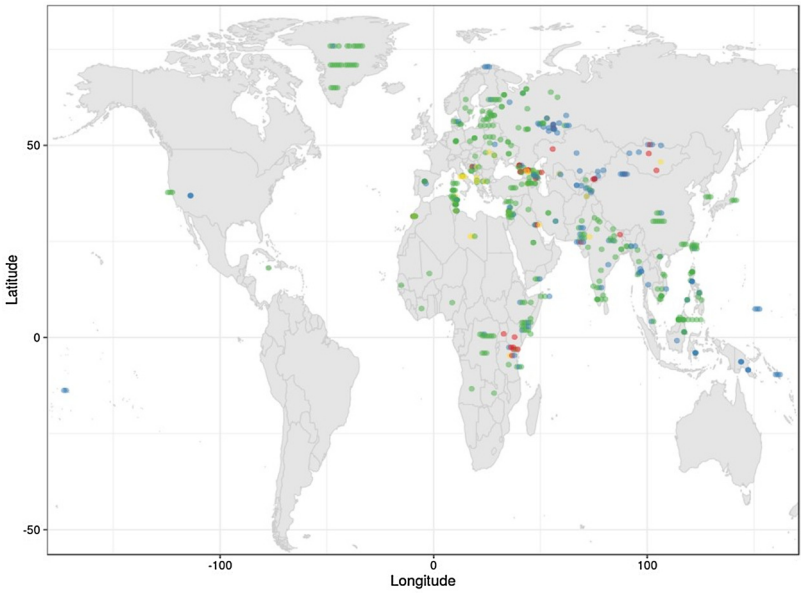


Fig. 2. Geographical position and assignment with *GenoGeographer* of 566 test individuals. Green dots: Concordant population assigned. Red dots: Discordant population assigned. Yellow dots: Ambiguous - more than one accepted reference population, with no statistically significant difference. Blue dots: No population assigned. For further details, see the legend to Table 1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

populations. Similar results were obtained with STRUCTURE (Fig. 4).

Fig. 4 shows a plot of the AIM profiles of the 3603 reference and 566 test individuals analysed with STRUCTURE and K = 4, considering the eight metapopulations (Fig. 4A). The test individuals were further divided into their known metapopulation of origin, and ordered within each group according to the four genetic clusters (Fig. 4B). When considering this grouping, the test individuals within each metapopulation were genetically more diverse than those included in the respective reference populations (especially Europeans, South/Central Asians, and East Asians). The analysis demonstrated that the investigated populations had four major genetic components (K = 4) characteristic of Sub-Saharan African, European, South/Central Asian, and East Asian populations. The Somalian population had mainly Sub-Saharan African, and European markers with a small contribution of South/Central Asian markers. The North African population had mainly European markers and a smaller contribution of Sub-Saharan African markers. The Middle East population had mainly European markers and a smaller contribution of South Central/Asian markers. The Greenlandic population has almost equal contributions of East Asian and European markers. The Sub-Saharan, European, South/Central Asian, and East Asian populations had mainly markers characteristic of their own populations.

Supplementary Figs. 3–6 show the principal component analyses (PCA) of the AIM profiles of the 3603 individuals in the eight reference populations and the 566 test individuals divided into no population assigned (rejected), concordant, ambiguous, and discordant assignment of population of origin.

The clustering of reference individuals and the position of the test individuals in the PCA highlights the differences between genetic and geographic classification when inferring ancestry. Although positioned within the cluster of the expected reference or close to this, 127 individuals were not assigned to any population of origin according to the z-score method (Supplementary Fig. S3). The positions of the concordant individuals were within the reference clusters (Supplementary Fig. S4). The positions of the ambiguous individuals were mainly between clusters or in the periphery of the clusters (Supplementary Fig. S5). The positions of the discordant individuals were different from the clusters to which they had been assigned (Supplementary Fig. S6). An example was an Moroccan individual, who had approximately 98% of Sub-Saharan African genetic ancestry and was positioned in the Sub-Saharan cluster. Other discordant individuals were positioned in the periphery of a neighbouring cluster, most likely reflecting mixed ancestry.

Supplementary Fig. 7 shows the distribution of the LR values for the AIM profiles of all test individuals in each of the eight reference populations. The individuals belonging to the East Asian, Sub-Saharan African, Somali, and Greenlandic reference populations were clearly separated from individuals from other populations by the LR values, while e.g. the LR distribution of the European test individuals was very similar to those of the Middle East, the North African, and the South/Central Asian populations. Supplementary Fig. 8 shows the mean squared deviations (Brier scores) of the STRUCTURE cluster membership coefficients, Q, between the reference means and each profile stratified by status: reference, concordant, discordant, ambiguous, and no population assigned. The deviations were smallest for concordant AIM profiles and larger for those that were discordant or without population assignment than those of reference and concordant AIM profiles. Due to the small number of AIM profiles in the group with ambiguous population assignment, the tendency was not clear.

4. Discussion

Identification of the population of origin of a human being by means of DNA investigations is of great importance in certain crime and identification cases. Many research groups and commercial companies are collecting information on genetic markers in a large number of populations investigated with various methods. Commercial companies are offering information on ancestry based on genetic investigations that constantly becomes more informative. Most forensic genetic laboratories are dependent on investigations with commercially available kits like the Precision ID Ancestry Panel (Thermo Fisher Scientific) for investigations of ancestry. For forensic genetic use, it is important to optimize and critically evaluate the efficiency of the ancestry prediction of the investigations. The Precision ID Ancestry Panel allows investigation of multiple AIM SNPs, in this case 165 SNPs. The kit is rather robust [23,24] and comes with an interpretation module that is used by many laboratories [44–46]. The interpretation is based on estimation of frequencies – or likelihoods – of the AIM SNP profile in a number of population data sets. The population in which the AIM profile has the highest likelihood is considered the population of origin. However, the software – like many other software – does not evaluate if the assigned population is relevant by a statistical test. Thus, a population completely irrelevant for the case may be assigned as the most likely population of origin. Another problem is the reliability of the

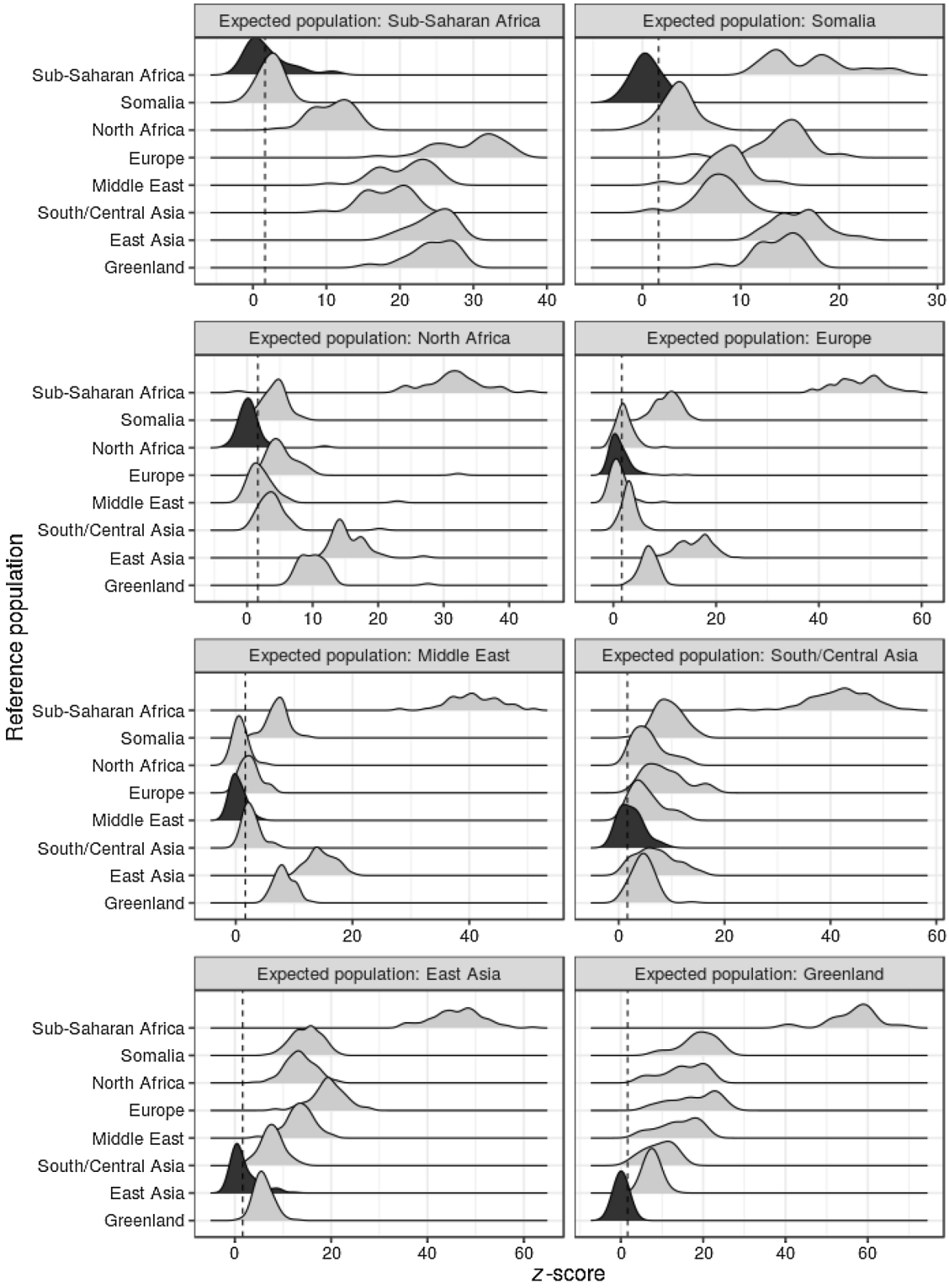


Fig. 3. Distributions of the z-scores of the AIM profiles of 566 test individuals.
Each line shows the smooth density histogram of the observed z-scores for each reference populations indicated to the left. Black colour indicates distributions of z-scores of the expected populations; grey colour indicates distributions of z-scores of unexpected population. The vertical, dashed lines indicate z-score = 1.64; P = 0.05 (one-sided test). Thus, data to the right of the dashed vertical lines suggest the AIM profiles to be rejected as members of the population indicated on the ordinate, while data to the left of the dashed, vertical lines represent AIM profiles that are *not* rejected as members.

discrimination between the various more or less likely populations of origin. Thus, the method is statistically naïve. We developed algorithms that offer help with handling these challenges [35]. The algorithms (cf. materials and methods) are included in the freely available software *GenoGeographer* (http://genogeographer.org/).

Due to lack of statistical methods based on likelihood principles that are recommended by the ISFG [33,34] for the evaluation of the possible affiliation of an AIM profile to a population of AIM profiles, we used the classical z-score test. Although the z-score test is not based on likelihood principles, we do not find any conflict with the ISFG recommendations

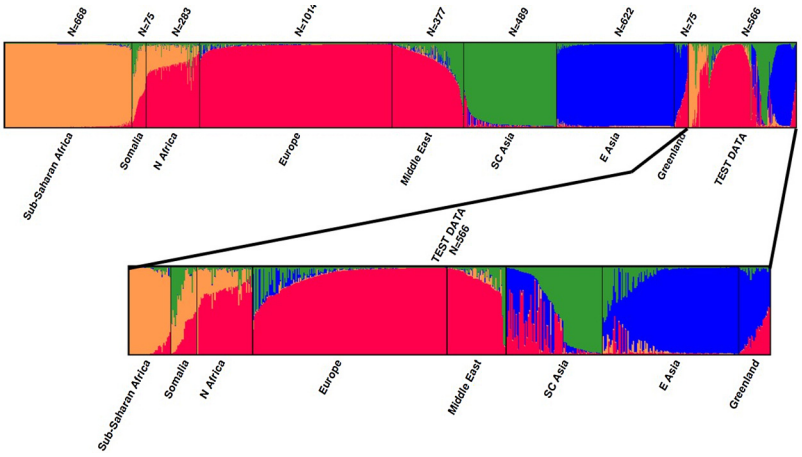


Fig. 4. Plot of membership coefficients obtained with STRUCTURE *2.3.4.21* for K = 4 using AIM profiles of the 3603 reference and the 566 test individuals grouped into the eight expected populations (cf. legend to Fig. 1). The test data were expanded in the lower part of the figure. The individuals were ordered according to the information retrieved on their population of origin.

that are focusing on the estimation of the weight of the evidence, while the z-score based test is used for evaluation of the relevance of the hypotheses of population affiliations. The significance level for assessing the results of the z-score test may be adjusted according to the circumstances of the case.

We compared the efficiencies of *GenoGeographer* and the naïve method. We did not use the software and data offered by Thermo Fisher Scientific. Instead, we established AIM SNP data that satisfied the need for sufficient numbers of individuals for each reference population, i.e. ≥ 75 individuals and evaluated the data with *Genogeographer* and the naïve method, respectively.

Forensic genetic laboratories have only made limited numbers of data on AIM SNP genotypes available. Therefore, we included SNP and DNA sequence data from other sources like the 1000 Genomes Project, the CEPH Panel, etc., that were collected and organized by Ken Kidd and colleagues, who kindly shared the data with us.

We included Greenlanders and Somalis in the investigations although both populations are admixed [47–49]. Our department serves Greenland that is part of the realm of the Kingdom of Denmark. Furthermore, a number of Greenlanders are living in Denmark. A significant number of Somali refugees were granted asylum in Denmark in the early 1990s. Together with their children and family members, who came to Denmark in connection with family reunification, they now form an important part of the population in Denmark. The analysis of the AIM SNP data of reference populations clearly showed that the Greenlanders and Somalis formed two distinct populations. Analysis of the test population showed that Greenlanders had the highest concordance rate of all investigated populations, and that Somalis had an average concordance rate. Thus, although the Greenlanders and Somalis come from admixed populations, the prediction of population origin is possible with *GenoGeogapher*.

The AIM SNP reference data are highly dependent on the quality of the validity of the information on population origin of the individuals as well as the SNP allele callings. Erroneous SNP allele calling is obviously a problem, but lack of calling of one of the alleles in heterozygous reference and test individuals also has a rather strong influence on ancestry prediction. The validity of SNP types in many accessible population databases is not at the same level as that in the forensic community. This is understandable since the purposes of population and forensic genetics are different. However, the issue must be taken into consideration when non-validated data from population studies are used for forensic genetic purposes.

The analysis of the reference population data with *GenoGeographer* showed that eight distinct populations could be defined. This was confirmed with STRUCTURE that identified four major genetic components (Sub-Saharan African, European, South/Central Asian, and East Asian) in the individuals in the eight populations.

The results of the test individuals showed that the rate of discordant population assignments was approximately three time less with *GenoGeographer* than with the naïve method. However, the rate of concordant assignments were higher with the naïve method than with *GenoGeographer* - the main reason being that 22.4% of the individuals were not (and should not be) assigned a population of origin, and 8.2% of the remaining individuals could belong to two or more populations when strict statistical criteria with *GenoGeographer* were used. The z-score distributions (Fig. 3) and the results of the *Structure* analysis indicated that a significant part of the test individuals may be of admixed ancestry. However, it is difficult to assess this possibility in detail, because it is impossible to revisit and interview the great majority of the individuals.

The lack of assignment of any population was highest among East Asian and South/Central Asian individuals, most like reflecting the limited amount of relevant data concerning these populations.

In forensic genetics, it is important to keep the error rate as low as possible and evaluate the results in a fair way. Thus, it seems better to admit that results are inconclusive than to make erroneous statements. It should, however, be noticed that the rather high error rates in this study most likely can be reduced significantly if the population affiliation of the investigated individuals is confirmed by meticulous questioning, and if typing and DNA sequencing is performed with validated forensic genetic methods by forensic genetic laboratories. The AIM SNPs in the Precision ID Ancestry Panel (Thermo Fisher Scientific) were selected in the beginning of the era of AIM SNP testing. A number of AIM SNPs that allow better discrimination between populations are constantly being identified. This will allow development of commercially available AIMs kits with better discrimination.

5. Conclusion

Based on own investigations and database information, we established a data set including the AIM SNPs in the Precision ID Ancestry Panel (Thermo Fisher Scientific) including 3603 individuals. Eight distinct populations (Sub-Saharan Africa, Somalia, North Africa, Europe, Middle East, South/Central Asia, East Asia, and Greenland) with four major genetic components (Sub-Saharan African, European, South/Central Asian, and East Asian) were identified. With *GenoGeographer* that includes a z-score based analysis for the relevance of assignment of populations, 77.6% of 566 AIM test profiles were assigned to a population, while 22.4% could not be assigned to any of the reference populations. The assignment of population was not concordant with the specified population of origin for 8.2% of the individuals when *GenoGeographer* used, and 21.9% when a naïve method based on the maximal likelihoods of the AIM profiles was used.

Declaration of Competing Interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.fsigen.2019.102154.

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